FITBUDDY

AI Fitness Plan Generator using Gemini Models

Date	01 October 2026
Team ID	SWTID-2026-2430
Project Name	Team Enrollment Confirmation – FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	2 Marks

Define Your Technology Stack

Identify the technologies used to build FitBuddy. The choices below align with the project requirements: HTML/CSS/Jinja2 pages, a FastAPI backend, Google Gemini models, and SQLite plan history. Hosting and version-control choices are suggested for the project and can be adjusted to match the team's implementation.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** User forms and plan display	HTML, CSS, Jinja2 templates	Creates clear input forms and readable results for weekly workouts, tips, and feedback. CSS supports responsive layouts for mobile and desktop.
2	**Backend / Server-Side** Application routes	Python, FastAPI	Handles form submission, validates inputs, coordinates plan generation and feedback updates, and exposes API documentation for testing.
3	**Database / Data Storage** Plan history	SQLite (SQLAlchemy if used)	Stores original and updated plans, user inputs, and feedback in a lightweight local database suitable for the project prototype.
4	**Cloud / Hosting / Deployment** Run environment	Local deployment; Docker optional	Local setup is simple for development and demonstration. Docker may be added to make the runtime more consistent across team members' machines.
5	**Version Control & CI/CD** Team collaboration	Git and GitHub	Tracks code changes, supports collaboration and review, and provides a shared repository. Automated CI can be added for repeatable tests.
6	**Third-Party APIs / Other Tools** AI generation	Google Gemini API; FastAPI interactive API docs	Gemini generates structured 7-day workout plans and concise nutrition/recovery tips, and can revise plans using user feedback. API docs support endpoint testing.

Note: FastAPI, Gemini models, HTML/CSS/Jinja2, and SQLite are specified in the FitBuddy solution requirements. GitHub and Docker are proposed supporting tools, not confirmed project requirements.

FITBUDDY — TEAM WORK ALLOCATION

Responsibilities are assigned in the same order as the team-member list provided.

S.No	Team Member	Assigned Responsibility	Expected Contribution
1	**Divya.D**	Requirements and coordination	Define user inputs, fitness goals, workflows, and requirements; coordinate and consolidate documentation.
2	**Deepika.M**	Gemini AI integration and prompts	Design prompts for structured 7-day workout plans, nutrition/recovery tips, and feedback-based revisions.
3	**Dharani.T**	FastAPI backend and routing	Implement form handling and routes for plan generation and feedback; connect routes to AI functions.
4	**Divya shree.R**	Frontend and user experience	Build HTML/Jinja2 pages for inputs, plans, tips, and feedback; style responsive layouts.
5	**DivyaDharshini.V**	Database, testing, and deployment support	Support SQLite/SQLAlchemy plan history, test main workflows, and verify local deployment and API documentation.

Shared team responsibility: All members review the final output, check that plans are clearly presented, and prepare the project demonstration.

Project scope: FitBuddy accepts basic user details and fitness goals, generates a personalized 7-day workout plan using Google Gemini models, provides a concise nutrition or recovery tip, supports feedback-based plan updates, stores plan history in SQLite, and provides an admin view.

Safety note: Fitness suggestions are general information and should be adapted to individual needs.